D1. GraphRAG vs Vector RAG, Refereed Properly

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TABLE I
QUESTION SET. $|G|$: MEAN GOLD DOCUMENTS PER QUESTION.

Type	Category	Used/All	$\|G\|$	#Q
Single-fact	Basic	50/175	1	50
Multi-hop	Intra-Document Reas.	25/40	1	50
	Project Related[a]	25/40	2.9	
Corpus-global	Completeness	20/20	6.5	50
	Project Related[a]	15/40	6.5	
	High Level[b]	10/10	0	
	Conflicting Info	5/20	2	
Fuzzy	Semantic	30/125	1	50
	Constrained	20/30	1.4	
Total				**200**

[a] All 40 used, split disjointly: the 25 with the fewest gold documents (2–4) are multi-hop, the other 15 (4–9) corpus-global. [b] No gold documents.

I. TASK / PROJECT IDEA

We implement GraphRAG and vector RAG over the same enterprise corpus and compare them per question type (single-fact, multi-hop, corpus-global, fuzzy). The deliverable is a *routing table*: for each type, which retriever should answer it and by how much it wins.

II. DATASET

We use EnterpriseRAG-Bench [1] (MIT license): ≈500k synthetic internal documents from 9 sources, includes deliberate noise (misfiled documents, conflicting near-duplicates) and internal codenames. Each of its 500 questions (10 categories) has a gold answer, a list of atomic *answer facts* (1–46 per question) and, except for *High Level* and *Info Not Found*, 1–10 gold document IDs.

Question set. We map categories onto the four Week 5 types and sample 200 questions, 50 per type, with a fixed seed (Table I); *Miscellaneous* and *Info Not Found* are not used.

III. METHOD / TECHNIQUE

Four retrievers: BM25, vector RAG with BGE embeddings, and GraphRAG, which extracts entities and relations from every chunk with an LLM, groups them into Leiden communities and summarises each. GraphRAG-*local* links question entities to the graph and expands their neighbourhood; GraphRAG-*global* combines community summaries. A rule-based router sends each question to the retriever the routing table assigns to its type.

IV. EVALUATION PLAN

Setup. All systems answer the 200 questions over the same index, with the same 512-token chunks, generator, prompt and context budget $B = 5 \times 512$ tokens, so only the retriever differs; BM25 is the baseline. Sample, prompts and judge are frozen before the first run.

Metric 1: supporting-fact coverage (Lab 5): $\mathrm{Cov}(q) = |R_q \cap G_q|/|G_q|$, where G_q are gold documents and R_q are the source documents of the context (for GraphRAG, provenance of its entities, relations and summaries). We report mean Cov, full-coverage rate and mean $|R_q|$ (exposes coverage bought with broad, many-source summaries).

Metric 2: answer correctness. A local LLM judge checks each gold answer fact against the answer; the score is the fraction of facts stated, and *fully correct* means all. Two of us label 40 answers (10 per type); we report human–human and judge–human Cohen's κ and grade manually any type with $\kappa < 0.6$.

Routing. Each type is assigned the retriever with the highest mean correctness, using stratified 2-fold cross-validation. Routers use gold or zero-shot LLM-predicted types (with classification accuracy) and are compared with the best single retriever and a per-question oracle. **Cost:** indexing GPU-hours and tokens; per-query latency, LLM calls and tokens.

Statistics and ablations. All metrics per type with 95% bootstrap CIs (1,000 resamples over questions); differences via paired bootstrap, and a CI containing zero is a tie, routed to the cheaper retriever. Ablations: local expansion depth 1 vs 2; $k \in \{3, 5, 10\}$ chunks.

Success: a per-type routing table with CIs, whichever retriever wins.

V. MAIN RISK

Graph construction over the full corpus. LLM extraction over ≈500k documents dominates cost and exceeds free Colab; it runs on our own GPU: NVIDIA GeForce RTX 5080, 32GB VRAM. We run it first, timing extraction on a 5k-document slice. *Fallback:* if the extrapolation exceeds budget, all systems use one fixed sub-corpus (gold documents of the 200 questions, their top BM25 neighbours, random documents from all sources).

REFERENCES

[1] Y. Sun, J. Rahmfeld, C. Weaver, R. Desai, W. Huang, and M. H. Butler, "EnterpriseRAG-Bench: A RAG benchmark for company internal knowledge," arXiv:2605.05253, 2026.

[2] Onyx, "EnterpriseRAG-Bench," GitHub repository. [Online]. Available: https://github.com/onyx-dot-app/EnterpriseRAG-Bench